

CN-TI-D2

Rezolvare: Răndu B

1. word addressable

64 bits/word

4 TiB virtual memory

32 GiB - physical memory

64 KiB page size

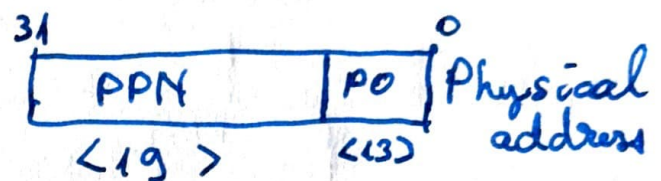
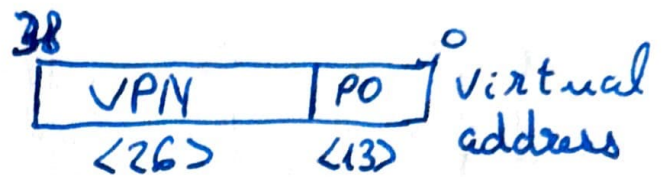
$$a) 4 \text{ TiB} = 2^2 \times 2^{40} \text{ B} = 2^{42} \text{ B}$$

$$32 \text{ GiB} = 2^5 \times 2^{30} \text{ B} = 2^{35} \text{ B}$$

$$64 \text{ KiB} = 2^6 \times 2^{10} \text{ B} = 2^{16} \text{ B}$$

$$64 \text{ bits/w} \Rightarrow 2^3 \text{ B/w}$$

$$\Rightarrow \text{VM} = \frac{2^{42} \text{ B}}{2^3 \text{ B/w}} = 2^{39} \text{ words}$$



$$\text{Page size} = 64 \text{ KiB} = \frac{2^{16} \text{ B}}{2^3 \text{ B/w}} = 2^{13} \text{ w} \Rightarrow \text{PO} = 13 \text{ bits} \text{ \& } \text{VPN} = 39 - 13 = 26 \text{ bits}$$

$$\text{PM} = 32 \text{ GiB} = \frac{2^{35} \text{ B}}{2^3 \text{ B/w}} = 2^{32} \text{ w}$$

Direct mapped TLB with 32 entries and $\frac{2 \text{ PTE}}{\text{block}}$

$\hookrightarrow 2^5 \text{ entries} \Rightarrow 5 \text{ bits index}$

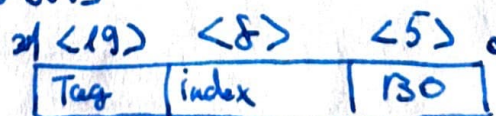
$\hookrightarrow \text{BO} = 1$

Cache - w. 32 w/block, 256 KiB data size, 4 way SA (LRU)

w B + w A

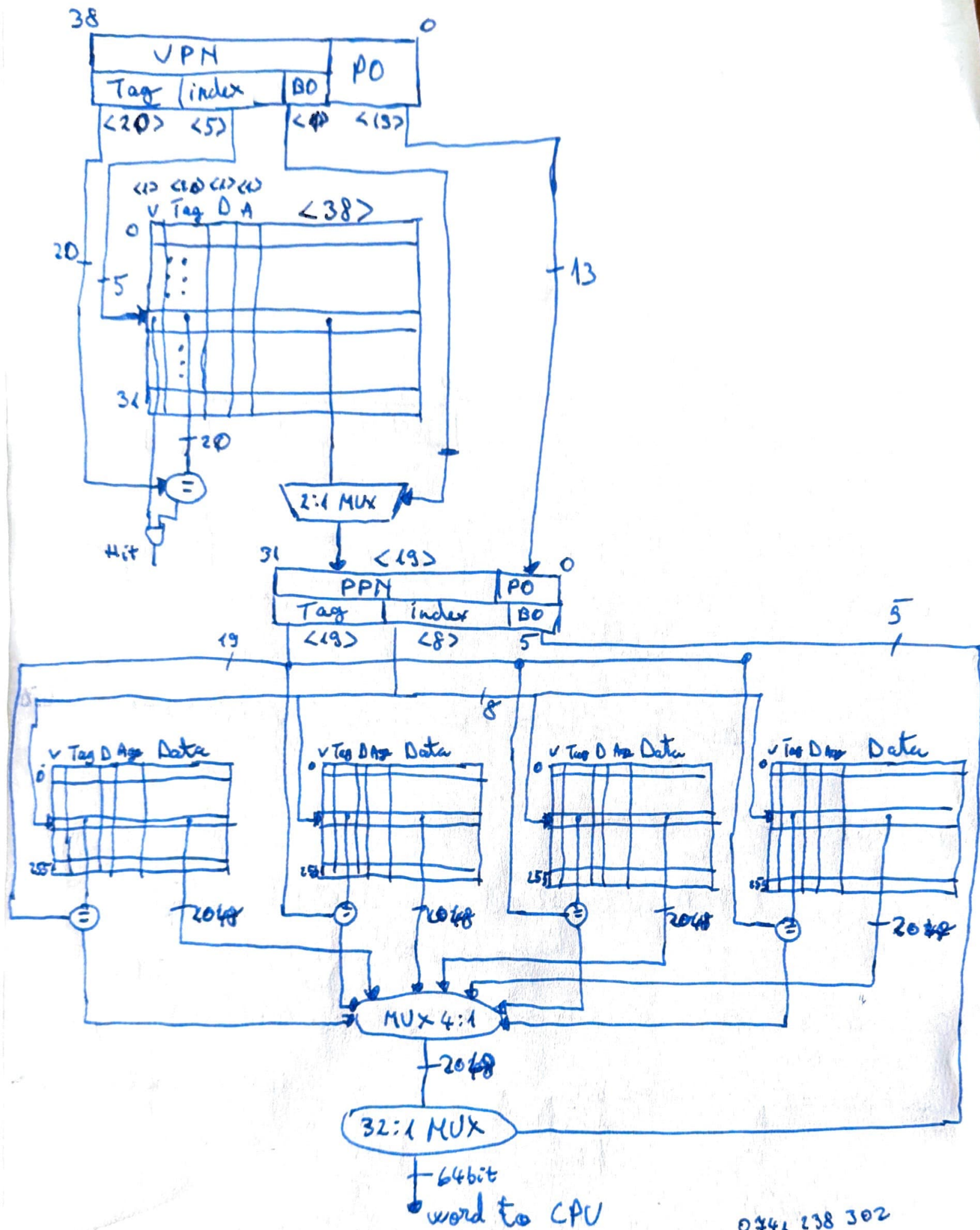
$$1 \text{ blk} = 32 \text{ w} = 2^5 \text{ w} \Rightarrow \text{BO} = 5 \text{ bits}$$

$$1 \text{ blk} = 2^5 \text{ w} \times 2^3 = 2^8 \text{ B/blk}$$



$$256 \text{ KiB} = 2^{18} \text{ B} = \frac{2^{18} \text{ B}}{2^{16} \text{ B/bank}} = 2^2 \text{ B/bank} \Rightarrow \frac{2^{16} \text{ B/bank}}{2^8 \text{ B/blk}} = 2^8 \text{ blk/bank}$$

c)



$$\text{LB total size} = 2^5 \times (1\text{bit valid} + 1\text{bit dirty} + 1\text{bit acc} + 20\text{bits tag} + 38\text{bits PPN})$$

$$= 2^5 \times 61\text{bits} \approx 2^5 \times 64\text{bits} \approx 2^5 \times 2^3 \text{B} = 2^8 \text{B} = \boxed{0.25 \text{KiB}}$$

$$\text{Cache total size} = 2^2 \times 2^8 \times (1\text{bit valid} + 19\text{bits tag} + 1\text{bit dirty} + 2\text{bits age} + 2^5 \times 2^3 \text{B})$$

$$= 2^{10} \times (23\text{bits} + 2^8 \text{B}) \approx 2^{10} \times 2^8 \text{B} + 2^{10} \times 2^2 \text{B} = 256 \text{KiB} + 4 \text{KiB} \\ \approx \boxed{260 \text{KiB}}$$

2. MOV R8, #0 ; used for array address MOV R3, #0 ; count = 0 MOV R4, #0 ; sum = 0 MOV R5, #0 ; i = 0 ADR R1, max ; get address for max ADR R2, N ; get address for N LDR R2, [R2] ; get value of N LDR R1, [R1] ; get value of max ADR R6, arr ; get address of base of arr .L1: LDR R7, [R6, R5] ; load arr[i] CMP R7, R1 ; arr[i] >= max BGT .L4 .L2 AND R8, R7, #1 ; arr[i] % 2 or TST R7, #1 CMP R8, #0 BNE .L4 .L3 ADD R3, R3, #1 ; count ++ ADD R4, R4, R7 ; sum += arr[i] .L4 ADD R6, R6, #4 ; increment by adding one word to array index ADD R5, R5, #1 ; increment i; CMP R5, R2 BLT .L1	ADR R1, count STR R3, [R1] ADR R1, sum STR R4, [R1]
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